

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 11: Longitudinal Stability & CG Margins (Neutral Point)

Syllabus Source: CAE Oxford (pp. 385-424) | Pass Target: $\geq 90\%$ AviationExam

1. LONGITUDINAL EQUILIBRIUM: WING & TAILPLANE MOMENTS

Pitch stability (rotation about the lateral axis) is dominated by the interaction between the wing and horizontal tailplane:

- **Wing Pitching Moment:** Cambered wings generate a constant nose-down pitching moment about their Aerodynamic Center (M_{AC}). Center of Gravity (CG) is positioned forward of the wing center of lift, creating an additional nose-down couple.
- **Tailplane Download (Downforce):** To balance these nose-down couples in trimmed flight, the horizontal tailplane must generate a continuous DOWNWARD AERODYNAMIC FORCE (Tail Download)! This downward force reduces effective total lift and increases fuel burn, but provides pitch stability.
- **Longitudinal Dihedral:** The angle of incidence of the horizontal stabilizer is set SMALLER than the wing incidence (tailplane incidence $<$ wing incidence). When the aircraft pitches up, the percentage increase in tailplane AOA is greater than the wing, generating an upward restoring lift that pitches the nose back down!

2. THE NEUTRAL POINT (NP) & STATIC MARGIN

The Neutral Point is the theoretical aerodynamic center of the ENTIRE AIRCRAFT (wing + fuselage + tail):

- **Neutral Point (NP):** The unique CG position where the overall aircraft pitching moment does not change with angle of attack ($dC_M / d\alpha = 0$). At the Neutral Point, static pitch stability is EXACTLY ZERO (neutral).
- **Static Margin:** The physical distance between the Center of Gravity and the Neutral Point, expressed as a percentage of the Mean Aerodynamic Chord (MAC): $\text{Static Margin} = (x_{NP} - x_{CG}) / MAC$.
- **Stable CG Regime:** For positive longitudinal stability, the Center of Gravity MUST BE LOCATED FORWARD of the Neutral Point! Static Margin must be POSITIVE (typically 5% to 15% MAC in civil transports).

CG Location Condition	Stability & Stick Force per 'g'	Performance, Drag & Stall Speed
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Forward CG Limit	MAXIMUM stability. Heavy stick forces; high stick force per g. Excellent recovery from stall.	High tailplane download required -> higher effective weight -> HIGHER STALL SPEED, higher drag, higher fuel burn. Limited by elevator authority in landing flare!
Aft CG Limit	REDUCED stability. Very light stick forces; low stick force per g. Highly sensitive controls.	Lower tailplane download -> lower trim drag, LOWER FUEL BURN, lower stall speed. Limited by minimum mandatory static margin!
CG Aft of Neutral Point	STATICALLY UNSTABLE! Any pitch disturbance causes divergent pitch runaway into deep stall or structural overload!	STRICTLY PROHIBITED in certified civil transport aircraft!

[!] EXAM TRAP (AVIATIONEXAM) - Forward CG vs Aft CG Flight Characteristics Trap

- FORWARD CG: More stable, heavier stick forces, higher stall speed, higher fuel consumption, flare authority limited! - AFT CG: Less stable, lighter stick forces (danger of overstressing airframe), lower stall speed, lower fuel consumption (higher cruise range)! - Aft CG Limit Rule: The aft CG limit is determined primarily by the minimum acceptable static longitudinal stability margin!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): CG Position Impact on Cruise & S**

Rule 'Aft = Aerodynamic efficiency, Forward = Forgiving stability': Step 1 (Aft CG): Shorter arm from weight to wing CP -> Less downward tail load needed -> Wing produces less total lift -> Lower induced drag -> LOWER fuel burn and HIGHER cruise speed. Step 2 (Forward CG): Long arm -> Large downward tail load -> Wing must produce extra lift -> Higher induced drag -> Higher fuel burn, but high static margin and easy stall recovery.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What happens to aircraft static longitudinal stability and stick forces as the Center of Gravity moves progressively aft?**

- [A]: Stability increases and stick forces increase
- [B]: Stability DECREASES and stick force per 'g' DECREASES (controls feel lighter and more sensitive)
- [C]: Stability remains unaffected
- [D]: The aircraft becomes uncontrollable immediately

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Moving CG aft shortens the distance to the Neutral Point (reducing Static Margin). This weakens restoring moments, reducing pitch stability and lowering stick force per g, making the aircraft more agile but sensitive.

Q2: Why does an aircraft cruising with a FORWARD Center of Gravity burn more fuel than with an AFT Center of Gravity?

- [A]: Engine thrust vector is misaligned
- [B]: Forward CG requires a larger downward tailplane force to balance the nose-down moment, increasing total lift required and induced trim drag
- [C]: Fuel cannot flow to engines
- [D]: Forward CG increases skin friction

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Forward CG creates a large nose-down moment, requiring a strong downward tailplane aerodynamic force. The wings must produce extra lift to support both aircraft weight and this tail download, increasing induced drag and fuel burn.

Q3: What aerodynamic position represents the absolute maximum aft theoretical limit for positive static longitudinal stability?

- [A]: The wing leading edge
- [B]: The NEUTRAL POINT of the aircraft
- [C]: The 25% MAC point

- [D]: The elevator hinge line

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If the CG reaches the Neutral Point, the Static Margin becomes zero (neutral stability). Moving the CG aft of the Neutral Point results in negative static stability (pitch divergence).